CMSRN performance under increasing routine-pattern deviation

H6 (30 min) · full test set · training-fitted scenario thresholds

(a) Chengdu · inflow

	Low	Moderate	High	Extreme
HA	+85.1	+62.8	+20.2	+12.0
SVR	+51.0	+31.2	+11.9	+8.0
VAR	+59.2	+39.8	+17.1	+8.1
DCRNN	+60.8	+17.5	+4.9	+5.0
STGCN	-1.1	+1.0	+1.2	+2.7
GWNet	-1.0	+2.4	+2.2	+3.7
STNorm	-4.5	-1.2	-1.2	+0.4
DSTAGNN	+81.9	+58.5	+20.5	+5.3
PDFormer	+78.5	+64.1	+52.1	+58.4
STWave	-4.9	-2.0	-1.5	+0.4
PatchTST	+70.0	+12.5	+7.5	+9.8
TimeMixer	+71.1	+17.3	+10.5	+9.9
STDN	+85.5	+70.4	+32.6	+5.6
MetaDG	+84.7	+66.5	+33.1	+12.4
HyperD	-0.7	+0.7	+0.2	+3.3
RAST	-0.7	+1.8	+2.5	+4.6
CMSRN MAE	**0.161**	**0.344**	**0.705**	**1.340**
Valid n	59,917	129,101	182,079	128,448

(b) Chengdu · outflow

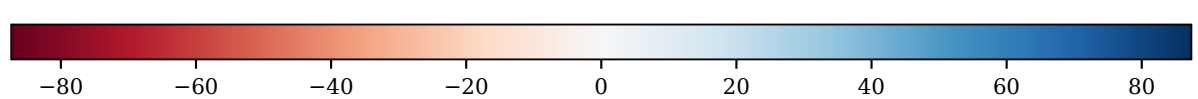
	Low	Moderate	High	Extreme
HA	+86.8	+58.5	+15.4	+9.0
SVR	+53.5	+25.7	+8.8	+5.4
VAR	+62.3	+33.2	+12.2	+6.0
DCRNN	+59.9	+12.1	+2.3	+2.1
STGCN	-1.2	-0.3	+0.2	+1.7
GWNet	-2.3	+0.4	+0.5	+1.9
STNorm	-2.8	-1.8	-1.5	-0.2
DSTAGNN	+84.8	+55.6	+18.5	+4.6
PDFormer	+80.3	+61.2	+50.5	+54.7
STWave	-3.1	-2.2	-1.2	-0.6
PatchTST	+72.3	+9.1	+5.1	+7.8
TimeMixer	+73.2	+14.0	+8.3	+8.4
STDN	+87.3	+66.3	+25.9	+4.3
MetaDG	+86.7	+63.2	+28.8	+10.2
HyperD	-1.1	-0.6	-0.3	+1.1
RAST	-0.9	+1.1	+1.4	+2.7
CMSRN MAE	**0.139**	**0.380**	**0.765**	**1.246**
Valid n	58,257	150,159	182,619	135,869

(c) Shanghai · inflow

	Low	Moderate	High	Extreme
HA	+82.7	+39.2	+12.3	+13.3
SVR	+50.1	+17.8	+8.4	+7.7
VAR	+60.5	+25.3	+10.8	+7.8
DCRNN	+10.1	+2.2	+1.9	+3.6
STGCN	+2.2	+1.2	+1.4	+4.0
GWNet	+1.2	+1.1	+1.6	+3.6
STNorm	-0.1	-1.1	-1.2	+0.5
DSTAGNN	+78.7	+33.0	+8.9	+5.1
PDFormer	+74.5	+52.5	+48.8	+57.7
STWave	-0.9	-1.0	-1.1	+0.2
PatchTST	+57.9	+6.4	+5.4	+10.2
TimeMixer	+58.9	+10.6	+8.2	+10.2
STDN	+84.6	+53.9	+25.6	+15.3
MetaDG	+83.1	+49.8	+24.8	+15.2
HyperD	+5.0	+0.8	+0.3	+2.8
RAST	+1.0	+0.9	+1.4	+3.1
CMSRN MAE	**0.177**	**0.536**	**0.851**	**1.296**
Valid n	92,199	223,697	222,486	115,210

(d) Shanghai · outflow

	Low	Moderate	High	Extreme
HA	+84.4	+37.9	+12.1	+12.1
SVR	+53.6	+16.5	+8.1	+7.2
VAR	+64.9	+23.6	+10.8	+7.4
DCRNN	+11.1	+2.1	+1.6	+3.4
STGCN	+2.1	+0.7	+1.1	+3.0
GWNet	+0.8	+0.6	+1.1	+3.6
STNorm	+0.1	-1.2	-1.2	+0.4
DSTAGNN	+81.2	+32.1	+9.3	+3.6
PDFormer	+76.7	+52.2	+49.9	+58.8
STWave	-0.3	-1.1	-1.0	+0.2
PatchTST	+61.5	+6.0	+5.1	+9.5
TimeMixer	+62.2	+9.8	+8.1	+9.7
STDN	+85.7	+43.0	+14.0	+6.1
MetaDG	+84.8	+49.6	+28.0	+16.6
HyperD	+1.9	+0.4	+0.0	+1.1
RAST	+2.3	+1.2	+1.5	+4.1
CMSRN MAE	**0.158**	**0.553**	**0.870**	**1.480**
Valid n	84,381	216,920	227,734	110,618



CMSRN MAE improvement over each baseline (%)

Improvement = 100 × (MAEbaseline − MAECMSRN) / MAEbaseline. Blue: CMSRN better; red: baseline better.

All finite values are shown without clipping; gray cells indicate an empty or undefined group.

Chengdu: CMSRN performance by routine-pattern deviation

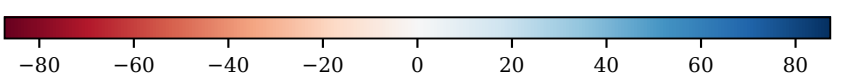
H6 (30 min) · full test set · training-fitted thresholds

(a) Inflow

	Low	Moderate	High	Extreme
HA	+85.1	+62.8	+20.2	+12.0
SVR	+51.0	+31.2	+11.9	+8.0
VAR	+59.2	+39.8	+17.1	+8.1
DCRNN	+60.8	+17.5	+4.9	+5.0
STGCN	-1.1	+1.0	+1.2	+2.7
GWNet	-1.0	+2.4	+2.2	+3.7
STNorm	-4.5	-1.2	-1.2	+0.4
DSTAGNN	+81.9	+58.5	+20.5	+5.3
PDFormer	+78.5	+64.1	+52.1	+58.4
STWave	-4.9	-2.0	-1.5	+0.4
PatchTST	+70.0	+12.5	+7.5	+9.8
TimeMixer	+71.1	+17.3	+10.5	+9.9
STDN	+85.5	+70.4	+32.6	+5.6
MetaDG	+84.7	+66.5	+33.1	+12.4
HyperD	-0.7	+0.7	+0.2	+3.3
RAST	-0.7	+1.8	+2.5	+4.6
CMSRN MAE	**0.161**	**0.344**	**0.705**	**1.340**
Valid n	59,917	129,101	182,079	128,448

(b) Outflow

	Low	Moderate	High	Extreme
HA	+86.8	+58.5	+15.4	+9.0
SVR	+53.5	+25.7	+8.8	+5.4
VAR	+62.3	+33.2	+12.2	+6.0
DCRNN	+59.9	+12.1	+2.3	+2.1
STGCN	-1.2	-0.3	+0.2	+1.7
GWNet	-2.3	+0.4	+0.5	+1.9
STNorm	-2.8	-1.8	-1.5	-0.2
DSTAGNN	+84.8	+55.6	+18.5	+4.6
PDFormer	+80.3	+61.2	+50.5	+54.7
STWave	-3.1	-2.2	-1.2	-0.6
PatchTST	+72.3	+9.1	+5.1	+7.8
TimeMixer	+73.2	+14.0	+8.3	+8.4
STDN	+87.3	+66.3	+25.9	+4.3
MetaDG	+86.7	+63.2	+28.8	+10.2
HyperD	-1.1	-0.6	-0.3	+1.1
RAST	-0.9	+1.1	+1.4	+2.7
CMSRN MAE	**0.139**	**0.380**	**0.765**	**1.246**
Valid n	58,257	150,159	182,619	135,869



CMSRN MAE improvement over each baseline (%)

Improvement = 100 × (MAEbaseline − MAECMSRN) / MAEbaseline. Blue: CMSRN better; red: baseline better.

Identical color scale across cities and the four-panel figure; no finite values are clipped.

Shanghai: CMSRN performance by routine-pattern deviation

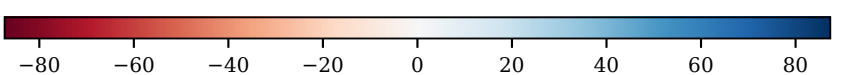
H6 (30 min) · full test set · training-fitted thresholds

(a) Inflow

	Low	Moderate	High	Extreme
HA	+82.7	+39.2	+12.3	+13.3
SVR	+50.1	+17.8	+8.4	+7.7
VAR	+60.5	+25.3	+10.8	+7.8
DCRNN	+10.1	+2.2	+1.9	+3.6
STGCN	+2.2	+1.2	+1.4	+4.0
GWNet	+1.2	+1.1	+1.6	+3.6
STNorm	-0.1	-1.1	-1.2	+0.5
DSTAGNN	+78.7	+33.0	+8.9	+5.1
PDFormer	+74.5	+52.5	+48.8	+57.7
STWave	-0.9	-1.0	-1.1	+0.2
PatchTST	+57.9	+6.4	+5.4	+10.2
TimeMixer	+58.9	+10.6	+8.2	+10.2
STDN	+84.6	+53.9	+25.6	+15.3
MetaDG	+83.1	+49.8	+24.8	+15.2
HyperD	+5.0	+0.8	+0.3	+2.8
RAST	+1.0	+0.9	+1.4	+3.1
CMSRN MAE	**0.177**	**0.536**	**0.851**	**1.296**
Valid n	92,199	223,697	222,486	115,210

(b) Outflow

	Low	Moderate	High	Extreme
HA	+84.4	+37.9	+12.1	+12.1
SVR	+53.6	+16.5	+8.1	+7.2
VAR	+64.9	+23.6	+10.8	+7.4
DCRNN	+11.1	+2.1	+1.6	+3.4
STGCN	+2.1	+0.7	+1.1	+3.0
GWNet	+0.8	+0.6	+1.1	+3.6
STNorm	+0.1	-1.2	-1.2	+0.4
DSTAGNN	+81.2	+32.1	+9.3	+3.6
PDFormer	+76.7	+52.2	+49.9	+58.8
STWave	-0.3	-1.1	-1.0	+0.2
PatchTST	+61.5	+6.0	+5.1	+9.5
TimeMixer	+62.2	+9.8	+8.1	+9.7
STDN	+85.7	+43.0	+14.0	+6.1
MetaDG	+84.8	+49.6	+28.0	+16.6
HyperD	+1.9	+0.4	+0.0	+1.1
RAST	+2.3	+1.2	+1.5	+4.1
CMSRN MAE	**0.158**	**0.553**	**0.870**	**1.480**
Valid n	84,381	216,920	227,734	110,618



CMSRN MAE improvement over each baseline (%)

Improvement = 100 × (MAEbaseline − MAECMSRN) / MAEbaseline. Blue: CMSRN better; red: baseline better.

Identical color scale across cities and the four-panel figure; no finite values are clipped.

Absolute forecasting error by routine-pattern deviation

H6 (30 min) · full test set · training-fitted scenario thresholds

(a) Chengdu · inflow

	Low	Moderate	High	Extreme
HA	1.08	0.93	0.88	1.52
SVR	0.33	0.50	0.80	1.46
VAR	0.40	0.57	0.85	1.46
DCRNN	0.41	0.42	0.74	1.41
STGCN	0.16	0.35	0.71	1.38
GWNet	0.16	0.35	0.72	1.39
STNorm	0.15	0.34	0.70	1.35
DSTAGNN	0.89	0.83	0.89	1.41
PDFormer	0.75	0.96	1.47	3.23
STWave	0.15	0.34	0.69	1.35
PatchTST	0.54	0.39	0.76	1.49
TimeMixer	0.56	0.42	0.79	1.49
STDN	1.11	1.16	1.05	1.42
MetaDG	1.05	1.03	1.05	1.53
HyperD	0.16	0.35	0.71	1.39
RAST	0.16	0.35	0.72	1.40
CMSRN	0.16	0.34	0.71	1.34

(b) Chengdu · outflow

	Low	Moderate	High	Extreme
HA	1.05	0.91	0.90	1.37
SVR	0.30	0.51	0.84	1.32
VAR	0.37	0.57	0.87	1.33
DCRNN	0.35	0.43	0.78	1.27
STGCN	0.14	0.38	0.77	1.27
GWNet	0.14	0.38	0.77	1.27
STNorm	0.13	0.37	0.75	1.24
DSTAGNN	0.91	0.86	0.94	1.31
PDFormer	0.70	0.98	1.54	2.75
STWave	0.13	0.37	0.76	1.24
PatchTST	0.50	0.42	0.81	1.35
TimeMixer	0.52	0.44	0.83	1.36
STDN	1.10	1.13	1.03	1.30
MetaDG	1.04	1.03	1.07	1.39
HyperD	0.14	0.38	0.76	1.26
RAST	0.14	0.38	0.78	1.28
CMSRN	0.14	0.38	0.76	1.25

(c) Shanghai · inflow

	Low	Moderate	High	Extreme
HA	1.02	0.88	0.97	1.49
SVR	0.36	0.65	0.93	1.40
VAR	0.45	0.72	0.95	1.41
DCRNN	0.20	0.55	0.87	1.34
STGCN	0.18	0.54	0.86	1.35
GWNet	0.18	0.54	0.86	1.34
STNorm	0.18	0.53	0.84	1.30
DSTAGNN	0.83	0.80	0.93	1.37
PDFormer	0.69	1.13	1.66	3.07
STWave	0.18	0.53	0.84	1.30
PatchTST	0.42	0.57	0.90	1.44
TimeMixer	0.43	0.60	0.93	1.44
STDN	1.15	1.16	1.14	1.53
MetaDG	1.05	1.07	1.13	1.53
HyperD	0.19	0.54	0.85	1.33
RAST	0.18	0.54	0.86	1.34
CMSRN	0.18	0.54	0.85	1.30

(d) Shanghai · outflow

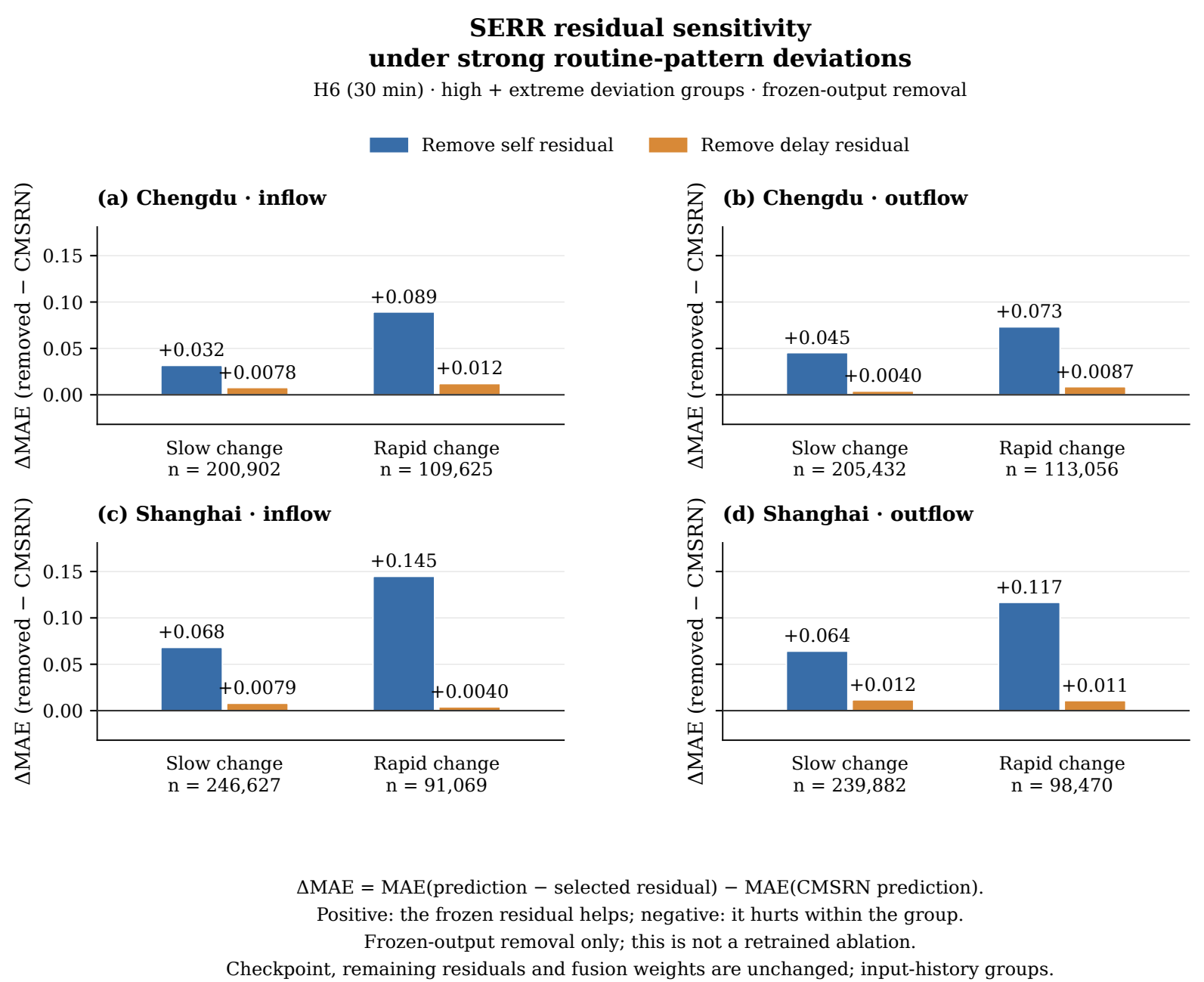
	Low	Moderate	High	Extreme
HA	1.01	0.89	0.99	1.68
SVR	0.34	0.66	0.95	1.59
VAR	0.45	0.72	0.97	1.60
DCRNN	0.18	0.56	0.88	1.53
STGCN	0.16	0.56	0.88	1.53
GWNet	0.16	0.56	0.88	1.53
STNorm	0.16	0.55	0.86	1.49
DSTAGNN	0.84	0.81	0.96	1.53
PDFormer	0.68	1.16	1.74	3.59
STWave	0.16	0.55	0.86	1.48
PatchTST	0.41	0.59	0.92	1.64
TimeMixer	0.42	0.61	0.95	1.64
STDN	1.10	0.97	1.01	1.58
MetaDG	1.03	1.10	1.21	1.77
HyperD	0.16	0.55	0.87	1.50
RAST	0.16	0.56	0.88	1.54
CMSRN	0.16	0.55	0.87	1.48



MAE (original flow units; lower is better)

Shared absolute-error color scale across all panels. CMSRN uses the existing Full checkpoint.

All finite values are shown without clipping; gray cells indicate an empty or undefined group.


SERR residual sensitivity
under strong routine-pattern deviations
H6 (30 min) · high + extreme deviation groups · frozen-output removal
Remove self residual
Remove delay residual
(a) Chengdu · inflow
(b) Chengdu · outflow
(c) Shanghai · inflow
(d) Shanghai · outflow
ΔMAE (removed − CMSRN)
0.15
0.10
0.05
0.00
+0.032
+0.0078
+0.089
+0.012
+0.045
+0.0040
+0.073
+0.0087
+0.068
+0.0079
+0.145
+0.0040
+0.064
+0.012
+0.117
+0.011
Slow change
n = 200,902
Rapid change
n = 109,625
Slow change
n = 205,432
Rapid change
n = 113,056
Slow change
n = 246,627
Rapid change
n = 91,069
Slow change
n = 239,882
Rapid change
n = 98,470
ΔMAE = MAE(prediction − selected residual) − MAE(CMSRN prediction).
Positive: the frozen residual helps; negative: it hurts within the group.
Frozen-output removal only; this is not a retrained ablation.
Checkpoint, remaining residuals and fusion weights are unchanged; input-history groups.

BCMS residual sensitivity: background difference and forecast horizon

High + extreme deviation groups · frozen-output removal

Candidate-background difference

(a) Chengdu · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 79.0k–82.6k	+2.175	+2.355	+3.074	+2.352	+1.287	+1.918	+2.001	+2.575	+2.859	+3.643	+2.639	+2.232
Medium n = 99.4k–102.1k	+1.405	+1.549	+2.075	+1.602	+0.752	+1.269	+1.346	+1.739	+1.945	+2.536	+1.784	+1.465
High n = 126.5k–130.1k	+0.639	+0.725	+1.051	+0.917	+0.381	+0.653	+0.645	+0.844	+0.949	+1.193	+0.839	+0.606

(b) Chengdu · outflow

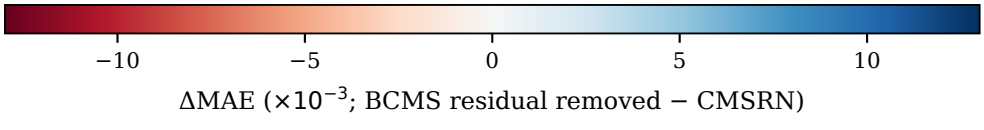
	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 85.8k–88.8k	+1.594	+1.316	+0.826	+1.089	+0.244	+0.481	+0.853	+1.419	+0.732	+2.018	+1.127	+0.483
Medium n = 111.3k–115.2k	+1.063	+0.911	+0.543	+0.617	+0.073	+0.184	+0.605	+0.891	+0.486	+1.430	+0.739	+0.256
High n = 114.9k–119.5k	+0.450	+0.386	+0.162	+0.233	-0.005	+0.069	+0.276	+0.435	+0.222	+0.725	+0.312	+0.115

(c) Shanghai · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 91.4k–95.1k	+12.350	+12.530	+12.671	+12.785	+12.213	+12.448	+12.315	+12.432	+13.002	+12.413	+12.700	+12.701
Medium n = 112.5k–115.9k	+8.895	+8.813	+9.068	+8.972	+8.626	+8.885	+8.782	+9.069	+9.342	+8.645	+9.120	+9.027
High n = 129.0k–131.0k	+6.166	+6.073	+6.212	+6.424	+6.295	+6.301	+6.151	+6.643	+6.855	+6.117	+6.562	+6.594

(d) Shanghai · outflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 90.6k–93.4k	+10.203	+9.899	+10.119	+10.149	+9.849	+9.809	+10.054	+9.990	+9.951	+9.914	+9.982	+10.146
Medium n = 110.7k–114.2k	+7.568	+7.404	+7.742	+7.645	+7.334	+7.452	+7.691	+7.601	+7.569	+7.396	+7.489	+7.561
High n = 131.0k–134.2k	+6.022	+5.793	+5.914	+5.841	+5.678	+5.771	+5.876	+5.601	+5.708	+5.541	+5.674	+5.523



ΔMAE (×10^{-3}; BCMS residual removed − CMSRN)

Values show 1000 × ΔMAE; divide by 1000 to recover original flow units.
ΔMAE = MAE(prediction − BCMS residual) − MAE(CMSRN prediction).
Frozen-output removal, not a retrained ablation or an isolated background test.
H1–H12: 5–60 min. Blue: helps; red: hurts. Counts span valid pairs across horizons.

BCMS residual sensitivity: background difference and forecast horizon

All test samples · frozen-output removal

Candidate-background difference

(a) Chengdu · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 221.8k–229.0k	+4.449	+4.659	+5.791	+4.394	+2.251	+3.450	+3.841	+4.549	+5.322	+6.802	+5.050	+4.235
Medium n = 135.9k–138.8k	+2.294	+2.494	+3.236	+2.415	+1.204	+1.925	+2.107	+2.604	+2.995	+3.894	+2.801	+2.343
High n = 135.3k–138.6k	+0.859	+0.972	+1.352	+1.141	+0.518	+0.829	+0.852	+1.081	+1.230	+1.554	+1.104	+0.838

(b) Chengdu · outflow

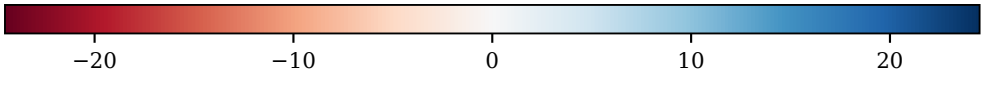
	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 244.6k–255.1k	+2.978	+2.525	+1.520	+2.291	+0.440	+0.767	+1.467	+2.799	+1.353	+3.830	+2.258	+1.090
Medium n = 149.8k–155.0k	+1.630	+1.398	+0.856	+1.108	+0.154	+0.331	+0.841	+1.431	+0.709	+2.162	+1.151	+0.503
High n = 122.2k–127.0k	+0.631	+0.547	+0.280	+0.388	+0.053	+0.148	+0.376	+0.590	+0.314	+0.936	+0.443	+0.216

(c) Shanghai · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 302.4k–307.7k	+23.924	+24.203	+24.480	+24.246	+23.824	+23.927	+23.944	+23.771	+24.502	+24.231	+24.507	+24.264
Medium n = 189.0k–192.8k	+14.355	+14.466	+14.805	+14.594	+14.147	+14.393	+14.297	+14.434	+14.891	+14.287	+14.758	+14.588
High n = 156.6k–158.7k	+8.564	+8.498	+8.680	+8.842	+8.646	+8.673	+8.553	+8.923	+9.181	+8.529	+8.951	+8.910

(d) Shanghai · outflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 293.9k–301.9k	+18.425	+18.107	+18.027	+18.434	+17.674	+18.145	+17.920	+18.359	+18.244	+18.406	+18.453	+18.875
Medium n = 182.4k–188.0k	+11.621	+11.424	+11.545	+11.611	+11.157	+11.343	+11.442	+11.557	+11.468	+11.403	+11.464	+11.643
High n = 153.8k–158.1k	+7.507	+7.292	+7.386	+7.332	+7.112	+7.258	+7.310	+7.123	+7.190	+7.061	+7.187	+7.139



ΔMAE (×10^{-3}; BCMS residual removed − CMSRN)

Values show 1000 × ΔMAE; divide by 1000 to recover original flow units.
ΔMAE = MAE(prediction − BCMS residual) − MAE(CMSRN prediction).
Frozen-output removal, not a retrained ablation or an isolated background test.
H1–H12: 5–60 min. Blue: helps; red: hurts. Counts span valid pairs across horizons.